Session 6: Make the numbers talk

Question: How do we show the finding so a manager understands it in five seconds?

Concepts

Concept	In daily life	In Excel	The limit
merge	Matching a packing list to the catalogue	VLOOKUP, XLOOKUP	Count rows before and after. A repeated key makes the table grow
matplotlib	Drawing a chart by hand, step by step	Insert Chart, then edit each part	Every chart needs a title and axis labels
seaborn	A chart template on top of matplotlib	Chart styles	Titles and labels work the same way

Code patterns

```
print(len(product_value))
product_value = product_value.merge(names, on="stock_code", how="left")
print(len(product_value))   # same number?


plt.figure(figsize=(10, 5))
plt.barh(top10["description"], top10["revenue"])
plt.title("The finding, in words")
plt.xlabel("Revenue (pounds)")
plt.ylabel("Product")
plt.tight_layout()
plt.show()

plt.xticks(rotation=45)   # when labels are long
```

Three common errors

Error	What to do
Table has more rows after a merge	A key repeats on the right. Use drop_duplicates(subset="stock_code") on the names table.
Chart shows twice	You called plt.show() two times. Call it once at the end.
Labels are cut off	Add plt.tight_layout() before plt.show().

Exercises and expected answers

Exercise	Expected
Merge check	3,066 rows, then 3,107, then 3,066 after the fix
Top 3 with names	REGENCY CAKESTAND 3 TIER, WHITE HANGING HEART T-LIGHT HOLDER, CHILLI LIGHTS
Fix the chart with no title	A finding-title and both axis labels
Seaborn (homework)	Top 5 outside the UK: Netherlands, EIRE, Germany, France, Australia

Case hunt: one page, five points

Point	Your answer
1. The problem	
2. Who is affected	
3. What data would show it	
4. One question you could answer with data	
5. Which tool fits: Python, SQL or a dashboard	